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in the device performance and reliability will be realised through a deeper understanding of the materials and basic physics of the device. The team in Bengaluru is tasked with building comprehensive models to help guide advanced device design.

Finally, GE's research team in Niskayuna, NY, is working on all facets of CdTe module development, including material growth, device development and robust process development. Their technical expertise cuts across diverse fields such as surface chemistry, laser processing and plasma physics. Equally diverse is their product development experience, which they have derived from GE's other technology-focused businesses such as healthcare, lighting and batteries.

In the future, the key areas where you will see improvements are efficiency and cost. These are the main factors influencing the future viability of the clean energy source.

What exactly is the nature of GE's relationship with PrimeStar? Have you licenced their CdTe thin-film technology, or is there some other sort of a relationship?

GE is a majority stakeholder of PrimeStar Solar, an emerging solar thin-film technology and manufacturing company.

What factors determine the efficiency of a solar panel?

The major factors include the amount of light that can pass through the window layers and reach the absorbing materials, the effectiveness of the absorber material to transform the incoming photons into carriers, and the ability to extract the carriers from the module with low losses.

How efficient are they in producing energy and also in terms of cost?

The market-leading thin-film modules today are slightly over 11 per cent efficient with costs of around 84 cents/watt.

Why is there so much push to thin-film solar cells? How different are these from conventional silicon solar cells?

Thin-film solar cells generally offer a lower-cost solution than alternative technologies and have a long runway for additional growth going forward.

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in thin films are impacted by the same price and supply volatility as has been seen with silicon. In addition, silicon-based solar modules generally have higher manufacturing costs than thin-film products.

Do you think that thin-film solar cells will make solar power a truly viable alternative?

The key to making solar power more viable is reducing the cost of producing electricity from the solar-based technology. The closer the solar gets to grid parity, or on par with fossil fuels like coal, the more the viability.

We believe thin-film solar represents one of the most promising ways to achieve these cost goals. These will cost less to produce and install versus silicon-based cells. And while their efficiency levels are not as high as that of silicon today, we think that the further improvements we make in efficiency, balanced against cost, will represent the most attractive option for customers.

In comparison to other thin-film solar technologies, what impact does CdTe have on the device performance?

Of all the thin-film materials, PrimeStar's CdTe solar technology matched up best with GE's R&D capabilities. We're already seeing promising results in the labs. Enhancing our materials understanding and the device design will be key to delivering a product that is truly best-in-class. And with CdTe, there is more headroom for device efficiency than with other thin-film PV modules.

Cd is a toxic material, so the solar panels based on CdTe will not be RoHS-compliant. Going by that, how successful will these be?

GE and others like the US Department of Energy (DoE) have studied this aspect. The US DoE has found that use of CdTe PV modules does not present any risk to health or environment, and recycling the modules at the end of their life addresses environmental concerns, if any.

Cadmium used in the GE/PrimeStar module is a stable compound when bonded with telluride. In addition, the module is encapsulated between two layers of glass. The amount of cadmium in a thin-film solar panel is less than in a typical battery.

During their operation, CdTe modules do not produce any pollutant and, furthermore, by displacing fossil fuels, these offer great environmental benefits. GE has a comprehensive recycling plan, and will work with customers to reclaim the product after the life-cycle is complete.

What are your long-term goals in terms of cost and efficiency?

When we go to the market in early 2011, we want to be the industry leader. We see GE's investment in thin film as an opportunity to offer the most cost-competitive solar solutions to our customers. ●